Techniques and guidelines for anonymising data

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PRUK Workshop 'Enhancing longitudinal data sharing: Data management strategies and best practices'

25/11/2024

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Introduction

Anonymisation is a valuable tool that allows data to be shared, whilst preserving privacy. The process of anonymising data requires that identifiers (direct and indirect) are changed in some way, which sounds simple, but in reality can sometimes be a complex and confusing process. Every data source is different and a number of factors need to be considered. We will introduce different types of data, different types of identifiers, and different levels of disclosivity one can aim for – all of which are factors that need to be considered. We will look at the different techniques we can apply to both quantitative and qualitative data, and introduce some step-by-step guidance. We hope this session will give you a better understanding of what to look out for and why, along with the different techniques and options available.

Before we start looking at the detail, we need to mention that this content, while applicable more broadly, is primarily influenced by and aligned with current UK legislation as of March 2024.

It is crucial for those working in international contexts or with data from multiple regions to familiarise themselves with and adhere to the relevant legal frameworks and data protection regulations.

Personal data, special category data and personal information

Researchers must adhere to data protection requirements when managing or sharing personal data. In the UK, the Information Commissioner's Office (ICO) provides in-depth information about personal data, including definitions and considerations for pseudonymisation and anonymisation, within the scope of the UK General Data Protection Regulation and the Data Protection Act 2018.

Personal data is defined by the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018. In essence, personal data is information that relates to an identified or identifiable natural person, be it directly or indirectly, taking into account other information derived from published sources.

Special category data, also sometimes referred to as sensitive personal data, are defined in Article 9 (1) of the UK GDPR. These data require additional attention and protection due to their sensitivity. Special category data consists of data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data to uniquely identify a natural person, data concerning health, a person's sex life or sexual orientation.

Personal information is defined by the Statistics and Registration Service Act 2007 as information that relates to and identifies an individual (including a body corporate). Personal information, compared to personal data, takes into consideration corporate bodies.

1. Anonymisation steps

A simplified high-level approach can consist of the following three core steps.

Step 1: Find and assess identifiers

Step 2: Implement anonymisation techniques

Step 3: Review the data and re-assess any remaining disclosure risk

2. Types of identifiers

Based on the **personal data** and **personal information** definitions an individual or a body corporate can be identified either **directly** or **indirectly**; hence identifiers are classified as either direct or indirect.

Direct identifiers - information that directly identifies data subjects; examples include name, address, National Insurance number, NHS number, IP address, email address etc.

Indirect identifiers - information that in combination, may uniquely identify data subjects, this can potentially be linked to other sources of data (such as the electoral register); examples include sex, age, region, occupation, income, ethnicity, religious affiliation etc.

3. De-identification and anonymisation

The ICO provides in-depth information about personal data and guidance for de-identification and anonymisation within the legal framework. With the introduction of the (UK) GDPR a clearer distinction has been made between the two concepts and therefore it is worth noting the differences between the two definitions.

De-identification (sometimes referred to as pseudonymisation) of data refers to the process of removing or masking direct identifiers. This means that any identifiers that might directly lead to the re-identification of data subjects such as their names and their addresses are either replaced by pseudonyms (fake names), codes or removed. As only direct identifiers are removed the data remains identifiable indirectly, hence data protection legislation applies to any additional processing of the data.

Anonymisation refers to the process of ensuring that the risk of identifying a data subject is negligible. This usually involves more than de-identification and often requires that data be further altered or masked, generally by treating indirect identifiers. Anonymisation allows data to be shared ethically and legally while preserving confidentiality.

3.1 Effective anonymisation

The ICO introduce a key concept in ensuring the utility of data, and this is effective anonymisation. They provide guidance on determining whether information qualifies as

personal data or anonymous information, affecting the applicability of data protection laws. The assessment involves careful consideration of the risk of identifiability and the factors influencing it.

Data can be classified into several categories based on how identifiable individuals are.

- **Directly identifiable data**: This is deemed as personal data.
- **Indirectly identifiable data**: This also falls under personal data if the risk of identifiability is not sufficiently low.
- **Data unlikely to be identifiable**: The risk of identifying an individual is considered sufficiently low.
- **Data that is impossible to identify**: This represents genuinely anonymous information.

3.2 Determining identifiability

In **determining identifiability**, one must consider:

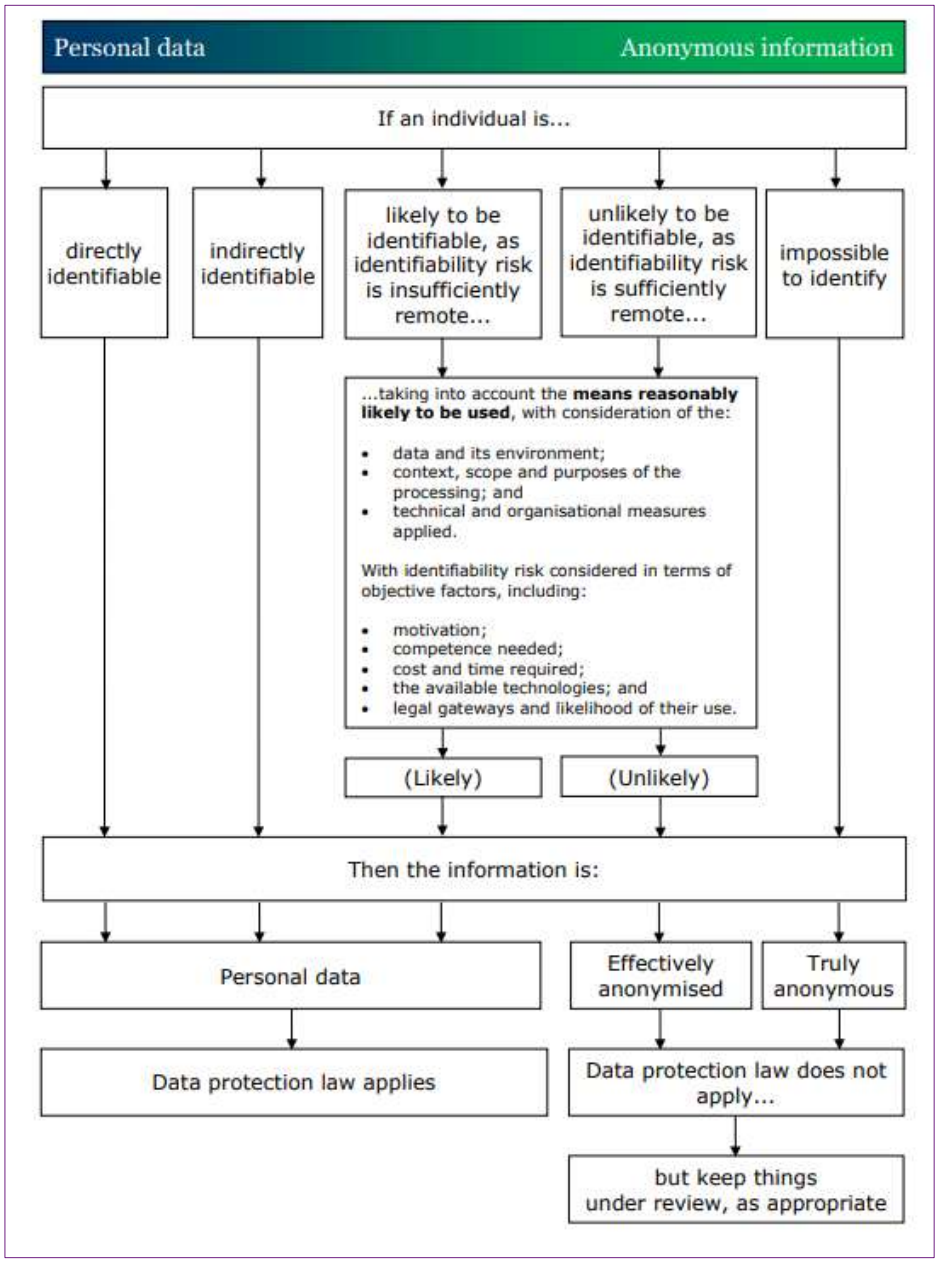
- The nature of the data and its environment.
- The context, scope, and purposes of its processing. Processing means taking any action with someone's personal data. This begins as soon as you have a record about someone and continues until you no longer need the information and it's been securely destroyed. If you hold information on someone, it counts as processing even if you don't do anything else with it (ICO).
- The technical and organisational measures in place.

The **risk is judged based on objective factors**, such as:

- The motivation to identify an individual.
- The level of skill required for identification.
- The cost and time investment needed.
- Available technological means.
- Legal avenues and the probability of their usage.

Considering the above, by using responsible repositories and making data available under appropriate licences, once anonymisation techniques have been applied, we can further ensure the data have been effectively anonymised.

3.3 Mapping the concept of the spectrum of identifiability to data protection law



Source: ICO (2021) [Chapter 2: How do we ensure anonymization is effective?]

Once you take all objective factors into account:

- if there are means "reasonably likely" to be used to identify someone, then you must view the information as personal data; and
- if no means are "reasonably likely" to be used, then you can view the information as effectively anonymised.

However, the identifiability risk must be sufficiently remote in the context of the processing.

Information that is effectively anonymised is not personal data and data protection law does not apply.

The feasibility and cost-effectiveness of the available means to identify individuals can change over time. In general, the more feasible and cost-effective a method becomes, the more likely it is to be a means reasonably likely to be used.

Additional safeguards could include terms and conditions of use. This will be covered in the final session of the day which looks at data sharing strategies and recommendations.

Activity: Quiz 1

Quiz 1 (Poll)

1. According to the UK General Data Protection Regulation, which of the following best describes 'personal data'?

A. Any data that is publicly available.

B. Data that relates to an identifiable natural person or body corporate.

C. Information relating to an identifiable natural person.

C
(Personal data is information relating specifically to an identifiable natural person. It does not include corporate information.)

Quiz 1 (Poll)

2. Which of the following is an example of 'Special Category Data' under the UK GDPR?

A. An individual's email address.

B. An individual's genetic data.

C. Corporate financial information.

B
(Genetic data is the only type of Special Category Data listed here. An individual's email address would be classed as personal data but is not sensitive personal data.)

Quiz 1 (Poll)

3. Does data protection legislation apply to de-identified data?

A. Data protection legislation applies to de-identified data because individuals can still be indirectly identified.

B. Data protection legislation does not apply to de-identified data.

A
(Data protection legislation applies because de-identified data, while having direct identifiers removed, can still potentially identify individuals indirectly.)

Quiz 1 (Poll)

4. What is the main difference between de-identification and anonymisation of data?

A. De-identification and anonymisation are the same processes.

B. De-identification involves removing direct identifiers, while anonymisation also involves altering indirect identifiers.

B

(De-identification involves removing or masking direct identifiers, whereas anonymisation also addresses indirect identifiers to ensure negligible re-identification risk.)

Quiz 1 (Poll)

5. Consider the following data excerpt from summaries created from a focus group discussion:

"Andrew Weaver, a 35-year-old engineer from Bristol, discussed his experiences with urban development projects."

Which of the following options correctly represents the de-identified version of this data?

A. Participant A, a 35-year-old engineer from Bristol, discussed his experiences with urban development projects.

B. A male participant discussed his experiences with urban development projects.

A

(The first option is the de-identified version of the data. It replaces the direct identifier, in this case the specific name of the participant 'Andrew Weaver' with a pseudonym 'Participant A', but retains other potentially identifying information like age, profession, and location.)

4. Anonymising quantitative data

We are now going to look specifically at the anonymisation of quantitative data. This might seem slightly more straightforward than anonymising qualitative data and arguably, it is. Nevertheless, there are different types of data and different types of identifiers we need to be aware of, as well as different levels of disclosivity we can aim for.

Let's look at **direct identifiers** in more detail. Direct identifiers may be collected as part of the research administration process but are usually not essential research information and can therefore easily be removed from the data. They should never be included in data made available for secondary analysis, unless explicit consent has been given by the respondents to share them. It is best practice to remove direct identifiers from any data which is being used for analysis. This ensures any identifying information is not copied multiple times and reduces the potential risk of disclosure. Examples include name, address, National Insurance number, NHS number, email address, driving licence number, social media handle but this list is not exhaustive.

Even if **direct identifiers** have been removed from the data, respondents may still be identifiable through indirect identifiers. These variables (sometimes referred to as key variables) may include information unique to the respondent, the household or business, which when used in combination with other indirect identifiers or linked with other information sources (such as the electoral register), could potentially identify data subjects.

Activity: Indirect identifier variables

Exercise

Please think specifically about the data you manage. Which details of any variables could be classed as an indirect identifier?

4.1 Indirect identifiers - context matters

The exercise has highlighted how many different possible indirect identifiers there are and how these can be unique to your data. We are going to look at indirect identifiers in more detail but we need to remember that selection of indirect identifiers should be considered in relation to the subject of your data and the sample characteristics.

For example, month of birth might be an important variable for a childhood study but this level of detail would not be appropriate a survey of working adults.

There might be cultural differences to take into account too. For example, a variable identifying arranged marriages from a survey conducted in Ethiopia might not be considered problematic because this is still common practice in Ethiopia, but identifying individuals who had an arranged marriage in England could lead to the identity of a participant being disclosed.

Similarly, a survey of a very specific group of people, for example women working at the UK Data Service, might mean that only limited demographic information can be included to reduce the risk of identification.

You need to balance data utility against potential information loss and consider how access controls can be used for mitigation.

4.2 Indirect identifiers - common examples

Identifying indirect identifiers in your data is an important step in the anonymisation process so we are going to spend some time looking at some of the common indirect identifiers in detail. This list is not exhaustive but will give you an idea of the types of variables to look for and how a combination of these variables can be used to uniquely point to someone. Categories are listed alphabetically rather than in order of importance, so we will start with age.

4.2.1 Age / date of birth variables

Age or date of birth is collected for the majority of surveys or individuals. It can be presented in a number of formats:

- Full date of birth
- Day of birth
- Month of birth
- Year of birth
- Age (single year)
- Age (in months)
- Age (banded)

Full date of birth should be only made available under Controlled Access conditions as can be used to link to other data sources. Where single age is available, you might find that small numbers of very young or old respondents are present in the data. This can be problematic as, when used in conjunction with other indirect identifiers, it can uniquely identify someone or be used to link to external data sources. Top-coding or bottom-coding might be required to group small numbers of very young or old respondents, or anonymisation of other variables in the file may make age/date of birth variables less disclosive.

	date_of_birth	day_of_birth	month_of_birth	year_of_birth	age_single_year	age_months	age_banded_5	age_banded_10
1	19.06.2000	19	6	2000	24	288	20-24 years old	20-29 years old
2	17.11.1980	17	11	1980	43	523	40-44 years old	40-49 years old
3	24.04.1931	24	4	1931	92	1115	80 years old and older	80 years old and older
4	10.01.2023	10	1	2023	1	17	0-4 years old	0-9 years old
5	15.08.1959	15	8	1959	64	778	60-64 years old	60-69 years old
6	26.07.1972	26	7	1972	51	622	50-54 years old	50-59 years old

Example: Age/ date of birth variables

Single year of age is often considered one of the most important demographic variables, therefore one has to keep in mind that banding and top-coding age can render some numeric analyses impossible (including mean age, for example), hence this needs to be considered when deciding for your course of action. Anonymisation of other variables in the file may make age/date of birth variables less disclosive and should be considered in conjunction with any aggregation.

4.2.2 Education variables

University names	Example university course names
Abertay University	Wildlife and Conservation Management
Aberystwyth University	Wildlife and Countryside Management
Anglia Ruskin University	Wildlife Conservation
Arts University Bournemouth	Wildlife Conservation and Ecology
Arts University Plymouth	Wildlife Conservation and Environmental Management
Aston University	Wildlife Conservation and Countryside Management
Bangor University	Wildlife Conservation Biology
Bath Spa University	Wildlife Ecology and Conservation
Birkbeck, University of London	Wildlife Ecology and Conservation Science
Birmingham City University	Wildlife Media

Example: Education variables

Education variables can be problematic when they include detailed information which could be unique or only applicable to a small number of the population. These variables, when used in combination with other variables, could lead to the identity of a participant being disclosed. School, college and university names may be disclosive, if analysed in conjunction with course titles or other information, especially where the data cover more than one course with accompanying institutional information. Education variables should also ideally be categorised using a coding frame. Where educational information is detailed, the usefulness of information for secondary analysis should be considered. Educational attainment or qualification level might be more useful than specific institutional information. For example the ONS Census 2021 'Highest level of qualification' variable codes education level into 8 categories. Removing some level of detail might considerably reduce the risk of identifying an individual.

4.2.3 Employment variables

Employment variables can be problematic when they include unique information or are only applicable to a small number of the population. Unique job titles (often in string variable format) may result in identification, especially, if other key variables in the data are very detailed. They can also include company names which would increase the risk of disclosure. String variable job titles should be removed, especially if the same information is categorised in other variables. Ideally employment variables should be categorised and coded, for example with SOC(2010/2020) or NS-SEC coding frames, though responses at more detailed levels of these schemas (four-digit SOC codes, for example), may also need aggregation (reduction to three or two digit codes – depending on the access level for the data), if frequency distributions are small.

Example SOC2020

Values	Categories
1	Employers in large organisations
2	Higher managerial occupations
3	Higher professional occupations
4	Lower professional and higher technical occupations
5	Lower managerial occupations
6	Higher supervisory occupations
7	Intermediate occupations
8	Employers in small organisations
9	Own account workers
10	Lower supervisory occupations
11	Lower technical occupations
12	Semi routine occupations
13	Routine occupations
14	Never worked and long-term unemployed
15	Full-time students
16	Not classified or inadequately stated
17	Not classifiable for other reasons
-9	Does not apply
-8	No answer

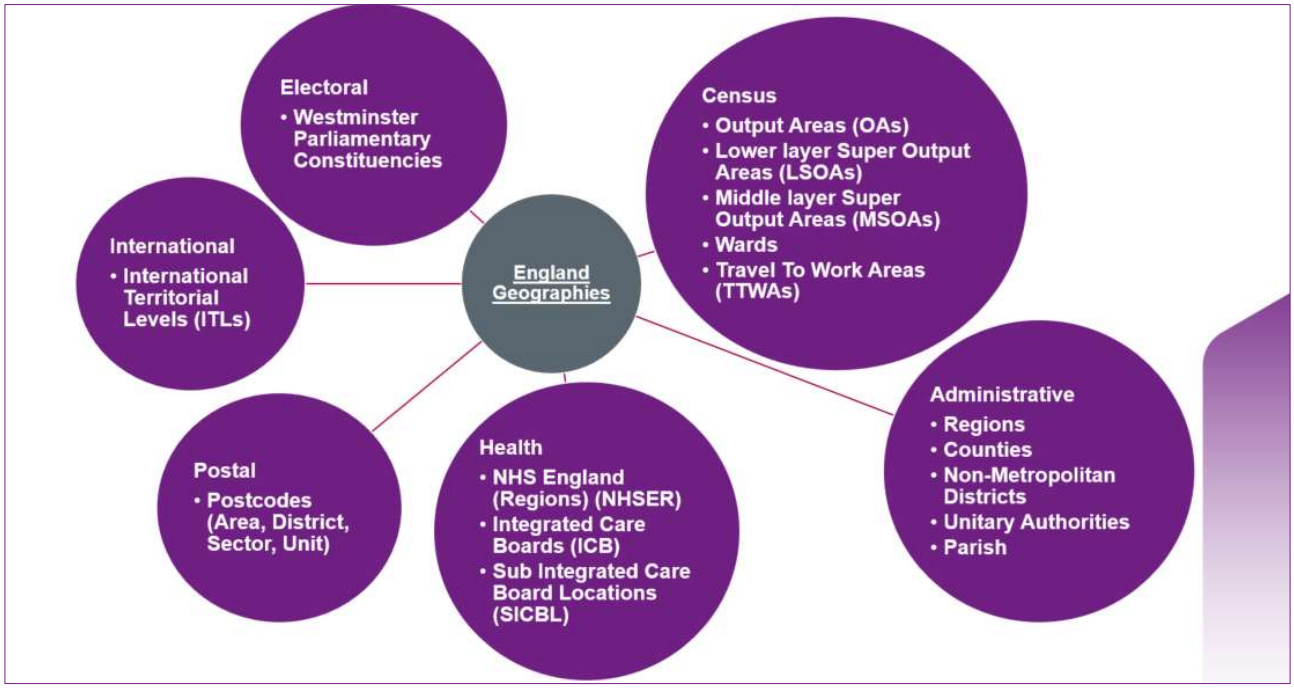
Example NS-SEC

Values	Categories
1	Managers, Directors And Senior Officials
2	Professional Occupations
3	Associate Professionaloccupations
4	Administrative And Secretarial Occupations
5	Skilled Trades Occupations
6	Caring, Leisure And Other Service Occupations
7	Sales And Customer Service Occupations
8	Process, Plant And Machine Operatives
9	Elementary Occupations
-9	Does not apply
-8	No answer

4.2.4 Ethnic group, national identity and religion variables

Detailed breakdowns of ethnicity, national identity or religious affiliation can be potentially problematic when used in combination with other variables. 'Other' responses which are often recorded in string format can include unique responses. Using a standard coding frame is helpful and the ONS provides some guidance on this complex area in Measuring equality: A guide for the collection and classification of ethnic group, national identity and religion data in the UK.

4.2.5 Geographic variables



Geographical or spatial variables present in the data should be considered carefully. Detailed, low level geographic variables will increase the ability to potentially identity data subjects. If the geographic variables are at a detailed level, it should be ensured that frequencies contain sufficient numbers to avoid identification in combination with other variables. For example, one case in a particular small area may mean that the respondent could be identified using a combination of other variables. (Some guidance may recommend cell counts, but the characteristics of the study, sample size and individual dataset should always be considered.)

The research topic of the study must also be borne in mind while checking geographical variables. For example in a politics-related study, low-level political geographies may be necessary for analysis, such as Ward or Parliamentary Constituency. Also, Scottish/Welsh/Northern Ireland surveys may include LA or similar level areas, as aggregation to Region/GOR level would mean that only country-level analysis would be possible (Wales, Scotland and NI are each one Region/GOR area).

Access level and geo-references are generally assessed together. Data available under Controlled Access conditions may also include other more detailed variables, in addition to lower level geographies.

4.2.6 Health information

Health information variables may be disclosive, if for example they contain unusual combinations of conditions or dates of hospital visits or named individual treatment centres. Aggregation, coding or removal of the variables affected may be the best solution.

height_cms	weight_kgs	date_of_appt	appt_location	health_cond1	health_cond2
168.20	110.50	19.04.2024	North Colchester Healthcare Centre	Diabetes	Asthma
131.70	62.60	22.05.2024	Elmstead Endoscopy Unit	Breast cancer	Cataract
180.50	50.30	01.04.2024	Oaks Hospital	Sickle cell anaemia	Epilepsy
151.50	100.20	15.04.2024	Colchester Hospital	Lung cancer	Diabetes
160.40	64.00	30.04.2024	Mary Barron suite	Stroke	High blood pressure
210.00	95.50	19.05.2024	UTC at Colchester Hospital	Heart attack	Arthritis

Example: Health information

4.2.7 Household size

Large households may mean that they are more unique and identifiable when their size is analysed alongside other variables. Where the survey is household based, the age and sex structure of the household members and relationships of individuals to each other can increase the risk of identification. Top-coding might be required depending on the other variables included and the individual study details. The ONS recommend top-coding household size at 10.

HHSize (D) Household size

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	1594	15.0	15.0	15.0
	2	3490	32.9	32.9	47.9
	3	2067	19.5	19.5	67.4
	4	2406	22.7	22.7	90.0
	5	719	6.8	6.8	96.8
	6	218	2.1	2.1	98.8
	7	61	.6	.6	99.4
	8	37	.3	.3	99.8
	9	9	.1	.1	99.8
	10 10 or more	16	.2	.2	100.0
	Total	10617	100.0	100.0	

Citation:
University of Manchester. Cathie Marsh Centre for Census and Survey Research. ESDS Government. (2004). *Health Survey for England, 2002: Teaching Dataset.* [data collection]. UK Data Service. SN: 5033, DOI: http://doi.org/10.5255/UKDA-SN-5033-1

RelToHRP Relationship to Household Reference Person

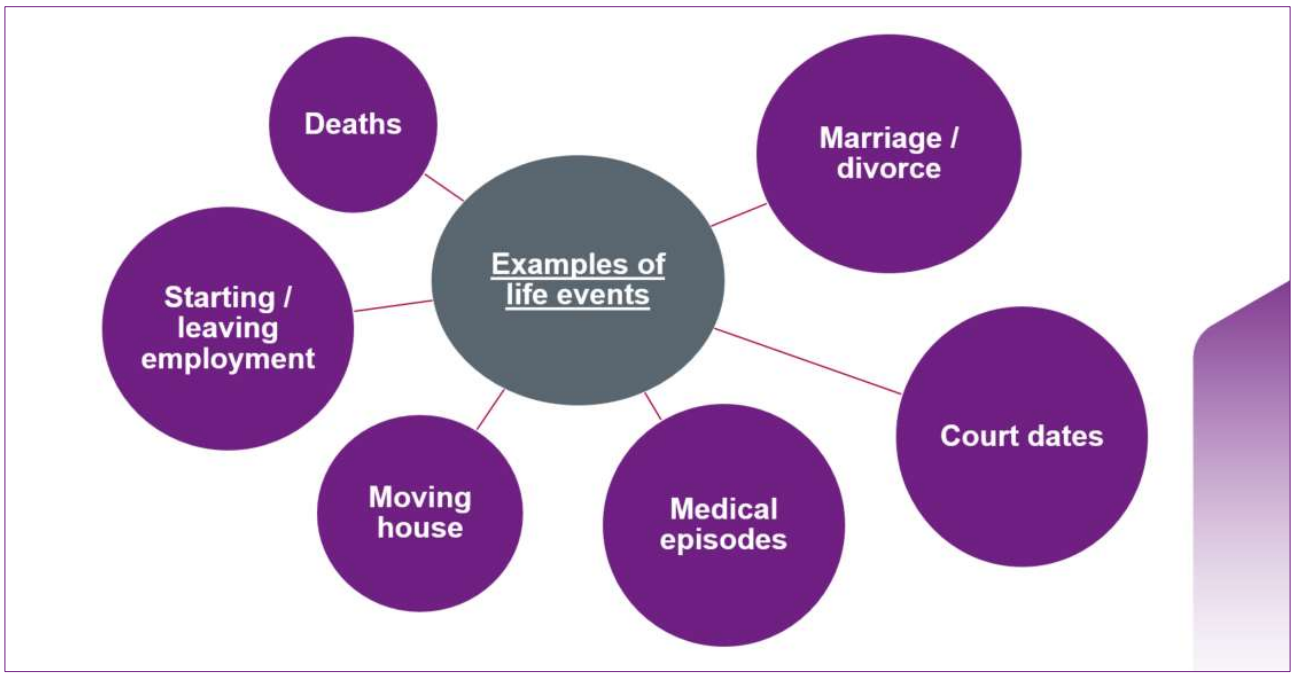
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1 Husband / Wife	2257	21.3	21.3	21.3
	2 Partner / Cohabitee	479	4.5	4.5	25.8
	3 Natural Son / Daughter	2554	24.1	24.1	49.8
	4 Adopted Son / Daughter	16	.2	.2	50.0
	5 Foster Child	6	.1	.1	50.0
	6 Step Son/Daughter / Child of Partner	73	.7	.7	50.7
	7 Son/Daughter-in-Law	12	.1	.1	50.8
	8 Natural Parent	41	.4	.4	51.2
	11 Step-Parent/Parent's Partner	1	.0	.0	51.2
	12 Parent-in-Law	6	.1	.1	51.3
	13 Natural Brother / Sister	23	.2	.2	51.5
	14 Half-Brother / Sister	1	.0	.0	51.5
	18 Brother / Sister-in-law	3	.0	.0	51.5
	19 Grandchild	47	.4	.4	52.0
	21 Other Relative	24	.2	.2	52.2
	22 Other Non-Relative	157	1.5	1.5	53.7
	96 Self	4917	46.3	46.3	100.0
	Total	10617	100.0	100.0	

Example: Household size and composition

4.2.8 Income and other financial variables

Income and other financial variables could be disclosive if they have unique outlying values. Isolated cases of a very high or low income or other financial sums – for example, a large lottery win recorded as unearned income during the time period the data cover - may present a disclosure risk if analysed alongside other variables and other publicly-available information (some lottery wins are well-publicised). Financial variables need to be checked and it may be advisable to recommend top-/bottom-coding to aggregate unique cases into a group. Banding and top-coding can hamper some statistical analyses where individual numbers are required, so data usability should always be considered.

4.2.8 Life events



Similar to date of birth variables, exact dates of life events (hospital treatments, court cases, marriages, deaths, etc.) may increase the risk of potential identification. Reducing exact dates to month and year might remove enough detail. Again it depends how crucial the variables are to the data usability so increased access conditions might be more appropriate.

As we mentioned earlier, this list is not exhaustive. These are just some of the common direct identifiers we encounter regularly. Each data source will be unique in terms of subject and sample.

4.2.8 Sensitive information

Data should always be checked for sensitive information. Some variables may include details that could be potentially harmful if a respondent is identified, such as an unusual or sensitive

health condition or status, or details of illegal behaviour. Some sensitive data will be Special Category data under GDPR and these variables should be considered alongside all other indirect identifiers in the dataset in case a disclosure risk is identified. They may need to be edited, but if doing so would harm the usability of the data, access restriction should also be considered.

The UK GDPR defines special category data as:

- personal data revealing **racial or ethnic origin**;
- personal data revealing **political opinions**;
- personal data revealing **religious or philosophical beliefs**;
- personal data revealing **trade union membership**;
- **genetic data**;
- **biometric data** (where used for identification purposes);
- data concerning **health**;
- data concerning a person's **sex life**; and
- data concerning a person's **sexual orientation**.

This does not include personal data about criminal allegations, proceedings or convictions, as separate rules apply. For further information, please see our separate guidance on criminal offence data.

ico. Information Commissioner's Office

Example: Sensitive personal data

4.2.9 String (text) variables

We have touched on string variables already but String (text) variables can include direct and/or indirect identifying information. These usually include the 'other' responses to survey questions so if the survey is self-completion the respondent can write whatever they want! It is amazing the level of detail that can be included in these variables. String variables should be checked thoroughly to ensure that they do not contain disclosive text (names, relatives' names, detailed locations, or anything else) that could result in the identification of respondents/organisations. String variables may need to be edited or removed.

- Other response/reason (please specify)

HOMP
{ASK IF HAS BEEN HOMELESS: HOMS = 1}
What was the main reason you moved out of the place you were living in before you became homeless?

INTERVIEWER: READ OUT

1 Tenancy came to an end
2 Home repossessed
3 Evicted
4 Fell out with parents
5 Parents' marriage/relationship broke down
6 Other reason (PLEASE SPECIFY) {VARNAME HOMO 200}

Example: String (text) variables

Activity: Risk considerations in longitudinal data

Exercise – Risk considerations in longitudinal data (1)

- Do you believe this information has been anonymised?
- Does any of the information look problematic?

	Wave 1	Wave 2	Wave 3
Age	18	20	22
Sex	Female	Female	Female
Marital status	Single	Married	Widowed
Household type	1 person	Cohabiting couple	1 person
Job title	In education	Police officer	Police officer
Region of current address	North East	North East	North East

Have a look at Example Table 1. The table shows data from one respondent collected over 3 waves of data collection. Imagine you have access to the data collection dates too.
- Do you believe this information has been anonymised?
- Does any of the information look problematic?

Considerations:
Wave 1 data looks o.k., but build-up of waves and change in circumstances could be problematic.
Change in marital status between waves is not common at such a young age.
The death of respondent's husband could be reported in the media, especially as he is likely to be young too.
The respondent is a police officer and is likely to wear a uniform so would be visually identifiable.
There is a chance that the husband, who died, could be a police officer too?
The geography is high level, so Wave 1 data is not problematic, but potentially unique changes in circumstances between waves could increase the risk of identification (especially, if husband's death is reported in the media).

Exercise – Risk considerations in longitudinal data (2a)

- Do you believe this information has been anonymised?
- Does any of the information look problematic?

	Wave 1	Wave 2	Wave 3
Age	99	101	103
Sex	Male	Male	Male
Marital status	Married	Widowed	Widowed
Household type	Cohabiting couple	Living in care	Living in care
Job title	Retired	Retired	Retired
Region of current address	East of England	East of England	East of England

Have a look at Example Table 2. The table shows data from a different respondent collected over 3 waves of data collection. Again, imagine you have access to the data collection dates too.
- Do you believe this information has been anonymised?
- Does any of the information look problematic?

Considerations:
Old age of respondent could be problematic at all waves.
Change in marital status between waves 1 and 2. Death of wife could be reported in media, especially, if too of an old age.
Change in household type between waves 1 and 2. Local people might know about a 101-year old widowed man moving house.
The geography is high level at region, which does reduce the risk of identification.

Now let us consider how introducing more detailed geography can change the risk. The next table shows the Example 2 we have just looked at, but with the addition of the Local Authority of the respondent's residence.
How does the addition of more detailed geography change the risk?

Considerations:
The change in LA between wave 1 and wave 2 corresponds with the respondent's change in household type when he moved into care.
We now know we are looking for a male around 100 years old who has moved into a care home in the Local Authority of Tendering. The introduction of more detailed geography dramatically increases the risk of identification.

Exercise – Exercise – Risk considerations in longitudinal data (2b)

- How does the addition of more detailed geography change the risk?

	Wave 1	Wave 2	Wave 3
Age	99	101	103
Sex	Male	Male	Male
Marital status	Married	Widowed	Widowed
Household type	Cohabiting couple	Living in care	Living in care
Job title	Retired	Retired	Retired
Region of current address	East of England	East of England	East of England
Local Authority of current address	Colchester	Tendering	Tendering

4.3 Extra considerations for anonymising longitudinal data

We also need to consider the complex nature of longitudinal data and extra considerations required when looking at anonymisation.

Any data which contains a longitudinal element pose a further risk, as linking successive waves or years can disclose more information about respondents than cross-sectional data. Each successive wave of data collection leads to an increase in knowledge about each individual or household. This means the risk of identifying a data subject naturally increases with each round of data collection.

Particular care should be given to:

- **Changes to individuals or household which can be linked to publically available information.** For example, a change in household size or marital status may increase the risk of identification of a household or individual by linking to births, deaths, marriage, civil partnership and divorce registrations, which are publically available.
- **Access to biomarkers overtime can easily create unique information.** For example, a change in a respondent's weight tracked over successive years could be identifying when used with other demographic details. A change in cortisol between waves might indicate the respondent has a rare condition called Cushing Syndrome. This also has physical symptoms such as a rounded face and pink and purple marks.

One simple solution to reduce the risk of disclosure would be to remove the ability to link individuals or households across successive years, but this would dramatically reduce the data usability. Other possible methods include higher level banding on demographic variables

such as ages, marital status, socio-economic and relationship variables, reducing the geo-reference variables available, along with access restrictions.

4.4 Statistical Disclosure Control (SDC)

Before talking about anonymisation techniques a wider concept must be introduced and this is Statistical Disclosure Control (SDC). SDC is a comprehensive approach to data privacy that goes beyond the traditional anonymisation techniques we will discuss in this introductory course and that applies both before and after data analysis.

Input SDC focuses on the raw individual-level data before any analysis or aggregation is performed. The goal here is to modify the data at the individual level to prevent the identification of data subjects in a structured and most often machine-aided manner. Techniques are often categorized based on whether they modify the data through perturbation or use non-perturbative methods to ensure the privacy of individuals in the dataset.

Perturbative methods involve altering the actual data values in some way to prevent the identification of individuals, while striving to preserve the overall statistical properties of the data. These include noise addition, data swapping, the Post Randomization Method, and synthetic data generation.

The **non-perturbative methods** do not completely change the actual data values but reduce the detail to prevent the identification of individuals. These include the techniques we will discuss shortly in more details such as recoding, banding, top/bottom coding.

There are several freeware that can be used for input SDC such as sdcMicro and the Amnesia Anonymization Tool.

Output SDC focuses on modifying the data after it has been **analysed or aggregated**. Output SDC is particularly relevant when publishing findings or releasing aggregated datasets. It ensures that the published information does not reveal sensitive details about individuals. Techniques in output SDC include:

- Table cell suppression: removing or modifying numbers in tabulated data that do not meet the SDC threshold.
- Controlled rounding: adjusting figures in tabular data to prevent back-calculation to exact numbers.
- Data swapping: similar to input SDC, but applied to aggregated data.

4.5 Anonymisation techniques

Once we have familiarised ourselves with the different types of identifiers present in quantitative data, we then need to decide on the best techniques to use to mitigate disclosure risk present in the data.

Before going ahead and making changes to data, remember that any changes should be made alongside data-sharing plans. Why? Because future access will dictate the level of anonymisation needed and we want to avoid under - or over - anonymisation. You need to consider how the data are going to be shared and the legal and ethical controls in place. This will influence the appropriate level of detail required to mitigate disclosure risk present in the data.

The most common quantitative anonymisation methods used are recoding, banding, top/bottom coding and generalisation. We will look at each in turn.

4.5.1 Recoding or Categorisation

We use this method to reduce the number of distinct categories or values of a variable, thus reducing disclosure risk.

Ethnicity
Black (Caribbean)
White (Irish)
Black (African)
Asian (Pakistani)
White (Scottish)
Black
White

Ethnicity
Black (Caribbean)
White (Irish)
Black (African)
Asian (Pakistani)
White (Scottish)
Black
White

4.5.2 Banding or Binning

Banding, also known as binning or grouping, is an effective anonymisation technique commonly applied to continuous variables like age or income. It involves categorising these variables into broader ranges or bands.

Income
£88,599
£21,478
£9,996
£51,299
£120,987
£197,000
£27,998

Income
£75,000 - £99,999
£10,000 - £24,999
Less than £9,999
£50,000 - £74,999
£100,000 or more
£100,000 or more
£25,000 - £49,999

4.5.3 Top/bottom coding

By top/bottom coding, the goal is to reduce disclosure risk added by small counts in the tails of a distribution, if there are concerns that there might be small counts. A good example here is age, where we might have some very small counts at either end of the spectrum (youngest and oldest research participants). By top and/or bottom coding we would recode, for instance, all under 10 or all over 80.

Age
27
118
89
56
48
31
5

Age
27
80 or older
80 or older
56
48
31
10 or younger

4.5.4 Generalisation

Survey data might contain free-text responses that increase the disclosure risk. To reduce the risk of disclosure you can generalise the meaning of a detailed text variable by replacing the potential disclosive information.

For example, in a dataset containing employee feedback, respondents provided detailed descriptions of their roles and responsibilities. These free-text responses could potentially be disclosive, especially in cases where the role is unique or highly specialised within the organisation.

The detailed descriptions of roles and responsibilities can be replaced with more general text or coded into broader categories. For instance, specific tasks and job responsibilities can be categorised under generic job functions.

- Original Detailed Response: "I am responsible for leading the cybersecurity initiatives, specifically focusing on blockchain technology security and data encryption techniques."
- Generalised Text or Coded Response: "Involved in IT security and data protection."

5. Anonymising qualitative data

Qualitative data poses particular challenges regarding anonymisation. Considerations need to be given to the level of anonymity required to meet the needs agreed upon during the informed consent process. Removing/redacting too much information reduces the value and integrity of the data, however, removing/redacting too little information might lead to potential identification of respondents.

5.1 Anonymising qualitative data - techniques

We are going to look at de-identification and anonymisation techniques in relation to direct and indirect identifiers found in qualitative data.

Use pseudonyms for direct identifiers

As already covered for quantitative data, direct identifiers refer to information which can be linked directly to the individual, such as a name or contact details. These should be disguised by allocating pseudonyms, or fake names/details, which should be consistent throughout the project, including in publications and follow-up research. Pseudonyms ensure the integrity of relationships inherent in the data and allow for a richer understanding of the data as a whole. Care should be taken to ensure that pseudonyms do not still signpost to participants in any way (e.g. using a nickname) and that names are changed in all files (including, for example, document metadata, interview transcripts, interview summaries, and data listings).

Categorising and generalising indirect identifiers

When anonymising transcript data it is crucial to handle indirect identifiers with care. In relation to qualitative data, indirect identifiers are pieces of information that, while not directly revealing an individual's identity, can potentially be combined with other data to lead to the identification of a person. These often include socio-demographic characteristics such as age, sex, occupation, education, location data or significant dates such as date of birth, date of employment.

Categorisation serves as an effective strategy to preserve the utility of data while better-preserving privacy. By grouping similar data into categories, researchers can maintain essential analytical value without exposing specific details. For instance, ages within the text can be classified into ranges such as [20-25] years, [26-30] years, etc., rather than making available exact ages. Similarly, occupations can be categorised based on an international standard classification, allowing for a broader but still meaningful analysis.

Generalisation is another technique where specific details are replaced with broader concepts. This approach can apply to various data types, including geographical locations, where specific places might be generalised to regions or countries. For example, generalising from "living in a small village near the town of Blackburn in Lancashire." to "living in a rural area in the North West of England".

5.2 Anonymising qualitative data - best practise

Anonymisation plans

Anonymisation plans outline the strategy in place for protecting participants' identities. The plan should outline what characteristics are automatically pseudonymised and what other information to look out for which may also be disclosive.

Use find-and-replace

Ensure all text is changed using tools like find-and-replace, to rule out the possibility of human error. Where relevant, be sure to explore style variations (capitalisation, punctuation, and pluralisation) of the text so that all possible permutations of the word are found. Text editor tools often have an advanced option to search for wildcards, or variations, of the text. Where there is no option for wildcards, you can also annotate this in your search using ? (for single characters) and * (for multiple characters).

Identify changes with brackets

Identify replacements in text clearly. A common way to highlight this is with [square brackets] or XML tags (e.g. <anonsec>anonymised text<anonsec>. Transcription guidelines should document which annotations denote changes in the text.

5.3 Anonymising qualitative data - open source tools

There are a number of open source tools available which can be helpful when anonymising qualitative data.

- The UK Data Service text anonymisation tool uses MS Word macros to help find disclosive information to remove or pseudonymise.
- The open-source text anonymisation tool Textwash uses Python to identify and replace direct identifiers, but can only run on text files (.txt), not in Word or RTF.
- FAMTAFOS is currently in development and is an open-source desktop app that utilises AI technology to anonymise text at scale. It is based on Textwash.

Most AI tools that can be used to anonymise text work on the principle of Named Entity Recogition (NER), and can be set to search for names, locations, occupations, etc. They will then tag them for subsequent human editing. Beyond the programs listed above, many new tools now offer free online facilities, but it is worth noting that uploading text into an online interface may risk the inadvertent sharing of unanonymised data. If using one of these online tools to anonymise qualitative data, you should first check with the website/tool owner that no text is retained within the system after use.

6. Intruder testing

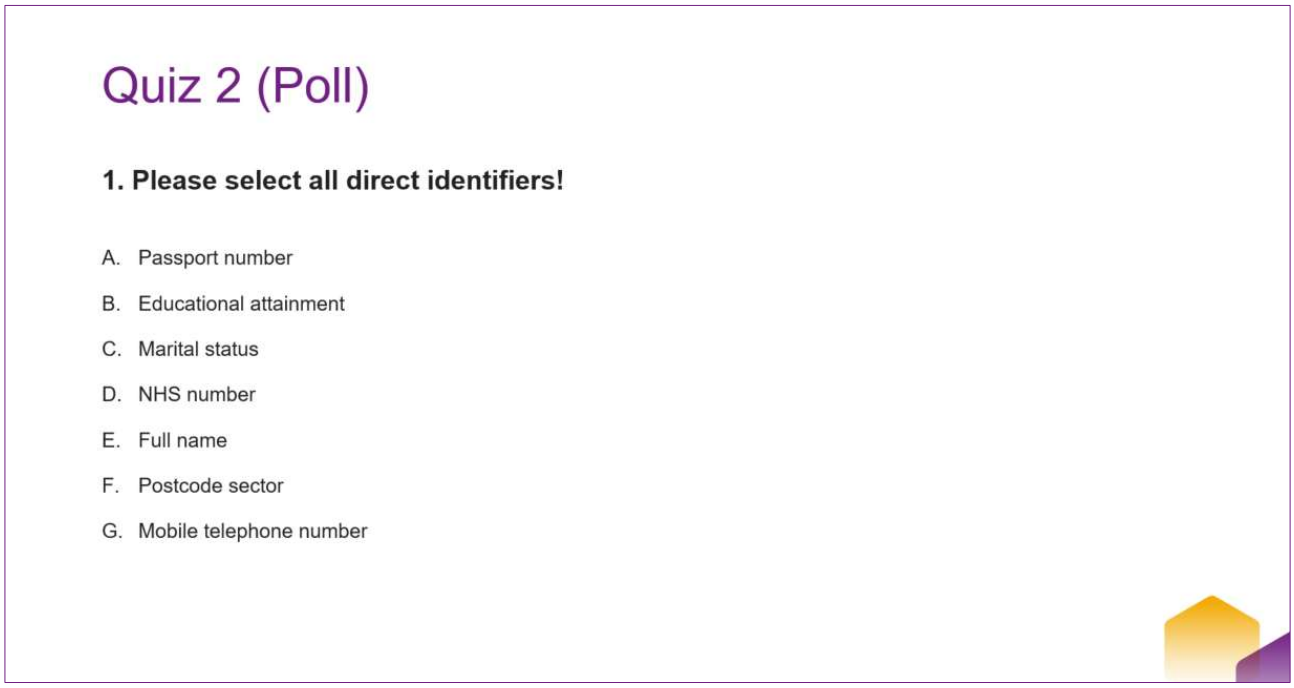
We discussed earlier how important it is to consider the motivation to identify an individual when assessing your data. You can take this a step further by completing Intruder Testing.

"Intruder testing involves using individuals described as "friendly intruders" to try and see if they are able to re-identify anyone in the dataset. These intruders should have some background knowledge of the data similar to that of a typical user. However, they do not need to be specialist hackers with the capability of employing advanced data exploration techniques." (ONS, 2024)

Intruder testing is not intended as a replacement for theoretical disclosure risk metrics, but a tool to be used alongside those more traditional methods.

The Office for National Statistics (ONS) provide further guidance and examples of how they have used this methodology to release microdata in their Guidance on intruder testing.

Activity: Quiz 2



A. Passport number (correct answer)
B. Educational attainment
C. Marital status
D. NHS number (correct answer)
E. Full name (correct answer)
F. Postcode sector
G. Mobile telephone number (correct answer)

Quiz 2 (Poll)

2. Please select all indirect identifiers, either sensitive or not sensitive!

A. Age
B. Educational attainment
C. Job title
D. Ethnicity
E. Religious affiliation
F. Household size
G. Life events e.g. exact date of marriage
H. Income
I. Life satisfaction rating
J. Health conditions
K. Internet usage (hours per day)
L. Air quality index
M. Sexual orientation

A. Age (correct answer)
B. Educational attainment (correct answer)
C. Job title (correct answer)
D. Ethnicity (correct answer)
E. Religious affiliation (correct answer)
F. Household size (correct answer)
G. Life events e.g. exact date of marriage (correct answer)
H. Income (correct answer)
I. Life satisfaction rating
J. Health conditions (correct answer)
K. Internet usage (hours per day)
L. Air quality index
M. Sexual orientation (correct answer)

Life satisfaction rating, internet usage and air quality index are unique to the respondent, the household or business.

Quiz 2 (Poll)

3. A data file contains 2 respondents with an ethnic group of *White: Gypsy or Irish Traveller*. Which anonymisation technique is most appropriate for reducing the risk of identification, if the data are to be released under Open Access?

A. Remove the ethnic group variable
B. Recode data into wider category
C. Banding/binning
D. Top/bottom coding

A. Remove the ethnic group variable
B. Recode data into wider category (correct answer)
C. Banding/binning
D. Top/bottom coding

Removing the whole ethnic group variable would unnecessarily remove too much data and reduce the utility of data. Banding/binning and top/bottom coding are techniques appropriate for continuous data such as age or income.

Quiz 2 (Poll)

4. Which of the following techniques are appropriate for anonymising qualitative data?

A. Replace direct identifiers with pseudonyms
B. Categorisation of indirect identifier
C. Generalisation of indirect identifiers
D. All of the above

A. Replace direct identifiers with pseudonyms
B. Categorisation of indirect identifier
C. Generalisation of indirect identifiers
D. All of the above (correct answer)

Direct identifiers should be replaced with pseudonyms and indirect identifiers should be masked using categorisation and generalisation to preserve the utility of data.

7. Summary anonymisation steps

Step 1: Find and assess identifiers

- Start by identifying potential identifiers. This should include both direct identifiers (information identifying data subject directly e.g. names, addresses) and indirect identifiers (information which when combined might identify data subjects e.g. age, sex, educational attainment, occupation).
- Evaluate the likelihood of re-identification by considering both the data itself and the potential availability of external information that could be linked to it.
- Key questions to consider:
    - Can the identity of a participant be deduced from information in the data file?
    - Is there a possibility of inadvertently disclosing or causing harm to a third party based on the information in the data file?

Step 2: Implement anonymisation techniques

- Ensure that all direct identifiers have been removed (deleted) or pseudonymised (replaced with fake names or codes).
- Next, address the indirect identifiers you have identified as potentially leading to identification. Techniques may include:
- Aggregation: Group data points to decrease identifiability. For example, rather than using specific ages, categorise them into broader age ranges such as 20-24, 25-29, 30-34, 35-39 etc.
- Generalisation: Modify detailed information to more general terms to prevent identification. Highly applicable for qualitative data such as transcripts, but also for survey data containing string variables. For example, generalise “living in the city of Preston in Lancashire” to “living in an urban location in the North West of England”.
- Data specific techniques: Incorporate specialised techniques such as recoding, top/bottom coding or statistical disclosure control in quantitative data, and methods like blurring or altering features in visual data, or voice distortion in audio data. Careful

consideration should be given to methods such as blurring or voice distortion; depending on the context and envisioned usage of the data, usability might be compromised.

- Key questions to consider:
  - How can the data be altered to prevent identification while retaining its utility for secondary analysis?
  - Are the anonymisation techniques employed sufficient to protect against re-identification? Make sure to consider the nature of the data including the data type and format, sensitivity, and uniqueness, as well as intended usage.

Step 3: Review the data and re-assess any remaining disclosure risk

- Ensure that the anonymisation process has been consistently applied across the data.
- Conduct a thorough review to confirm that no real residual risk of disclosing personal or sensitive information remains. If there is a low residual risk of disclosure, consider an effective anonymisation approach, a concept introduced by the ICO.
- Key questions to consider:
  - Have all identifiers, both direct and indirect, been adequately anonymised or removed?
  - Is there any remaining information that, when combined with other available data, could lead to the identification of individuals?

Activity: Practical anonymisation of survey data

Finally, there is the offer to complete an ‘Anonymisation of Survey Data Task’ outside the workshop/training using the provided fictional dummy dataset.

Task (to be completed outside of the workshop)

Exercise: Practical anonymisation of survey data

A dummy dataset (fictional data) has been created to facilitate this exercise.

Imagine a scenario where you must prepare the data for dissemination to secondary researchers from an extensive research initiative aiming to explore the intricate relationships between lifestyle choices, personal characteristics, and health outcomes across a diverse population. The study gathered comprehensive data from individuals from various regions to assess patterns in health behaviours, media consumption, travel habits, and socio-economic status.

Use your knowledge gained during the training sessions to assess the data for release under Safeguarded Access and Open Access.

Complete the provided table with details of the variables of concern, the potential disclosure risk, and suggested edits required for release under Safeguarded Access and Open Access.

To obtain the dataset for the Task, please download the PRUK training exercise and decide which data format you are going to use.

Task (to be completed outside of the workshop)

Exercise: Practical anonymisation of survey data

- When you click on the link https://ukdataservice.ac.uk/app/uploads/pruk_training_anon_exercise.zip, an automatic download of a zip bundle (pruk_training_anon_exercise.zip) starts. It contains the following files:

File name	Description
pruk_training_anon_exercise.docx	Exercise worksheet in Word format
pruk_training_anon_exercise.pdf	Exercise worksheet in PDF format
pruk_training_anon_exercise.csv	Data in CSV format
pruk_training_anon_exercise.dta	Data in Stata format
pruk_training_anon_exercise.sav	Data in SPSS format
pruk_training_anon_exercise.tab	Data in Tab-Delimited format
pruk_training_anon_exercise_ukda_data_dictionary.rtf	Data dictionary in RTF format
pruk_training_anon_exercise_answers.pdf	Exercise answers

- You can choose, which data format you wish to work in.

8. Resources

ICO - Chapter 2: How do we ensure anonymisation is effective?
ico.org.uk/media/about-the-ico/documents/4018606/chapter-2-anonymisation-draft.pdf

ICO - Anonymisation: managing data protection risk code of practice
ico.org.uk/media/1061/anonymisation-code.pdf

ICO - Key data protection terms you need to know: Processing
ico.org.uk/for-organisations/advice-for-small-organisations/key-data-protection-terms-you-need-to-know/#processing

ONS Policy for social survey microdata
www.ons.gov.uk/methodology/methodologytopicsandstatisticalconcepts/disclosurecontrol/policyforsocialsurveymicrodata

CESSDA Data Management Expert Guide
dmeg.cessda.eu/Data-Management-Expert-Guide/5.-Protect/Anonymisation

De-identification and Disclosure Review Procedures (CD249) (not to be shared externally):
ukdataservice.atlassian.net/wiki/spaces/SERVREPODATA/pages/33718295/De-identification+and+Disclosure+Review+Procedures+CD249

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This project is funded by the UKRI Economic and Social Research Council and the UKRI Medical Research Council.